

Nicholas J. Bryan, Ph.D.

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Principal Scientist & Manager

Web: <https://njb.github.io>

SUMMARY

- AI researcher & manager w/12+ years' experience audio R&D (3+ years as manager)
- 13+ tech transfers, 50+ papers, 14+ patents, 15+ patents pending, IEEE Senior member
- Interested in generative models (diffusion, GANs, LLMs), content authenticity

EDUCATION

Ph.D., Computer-based Music Theory and Acoustics, Stanford University.

M.S., Electrical Engineering, Stanford University.

M.A., Music, Science, and Technology, Stanford University.

B.S., Electrical Engineering., U. of Miami-FL, Summa cum laude.

B.M., Music Eng. Tech., U. of Miami-FL, Summa cum laude.

WORK EXPERIENCE

- **Adobe Research** (San Francisco, CA).....07/18-Now
 - Head of Music AI Group & Principal Scientist (05/2026 – Now)
 - Led 0-to-1 cross-org effort to ship generative music at Adobe
 - Shipped industry-first studio-quality, commercially safe GenAI music for *any* use.
 - Owner of all modeling and data work for Firefly Music Model
 - Head of Music AI Group & Senior Research Scientist II (12/23 – 05/2026)
 - Senior Research Scientist II (02/22 – 05/26)
 - Senior Research Scientist I (07/20 – 02/22)
 - Research Scientist II (07/18 – 07/20)
- **Apple Inc.**, Interactive Media Group (Cupertino, CA).....01/14-07/18
 - Senior Audio Algorithm Engineer (10/16 – 07/18)
 - Audio Algorithm Engineer (01/14 – 10/16)

TECHNOLOGY TRANSFERS

- Adobe Technologies
 - Generate Music (lead) – [Premiere Pro](#), 2026.
 - Generate Music (lead) – [Firefly Mobile](#), 2026.
 - Generate Music (lead) – [Firefly AI Assistant](#), 2026.
 - Beats (lead) – [Adobe Premiere Mobile](#), 2026.
 - Audio Match (lead) – [Adobe Stock AI Studio](#), 2026.
 - Generate Music (lead) – [Adobe Firefly \(web & mobile\)](#), 2025.

- Filler Word Removal (consult) – [Premiere Pro](#), 2023.
- Convertible Melspectrograms (lead) – [Premiere Pro](#), 2023.
- Voice Activity Detection (consult) – [Premiere Pro](#), 2023.
- AutoTone Lumetri Color (consult) – [Premiere Pro](#), 2022.
- AudioTagger (co-lead research) – [Adobe Project Blink](#), 2022.
- MicCheck (lead) – [Adobe Podcast](#), 2022.
- Find Similar Audio (lead) – [Adobe Stock](#), 2021.
- Sensei Toolkit (lead) – [Sensei Machine Learning Platform](#), 2020.
- Merge Clips (lead) – [Premiere Pro](#), 2013.
- PR Events: Three MAX Sneaks + 1 PR Event
- Apple – Single & multi-channel speech enhancement for iPhone + AirPods Pro 2014-2018.

JOURNAL PUBLICATIONS (PEER-REVIEWED)

7. Y. Bai, J. Casebeer, S. Sojoudi, N. J. Bryan, “DRAGON: Distributional Rewards Optimized Diffusion Generative Models.” *Transactions on Machine Learning Research (TMLR)*, 2025.
6. G. Zhu, J.P. Caceres, Z. Duan, N. J. Bryan, “MusicHiFi: Fast High-Fidelity Stereo Vocoding.” *IEEE Signal Processing Letters (SPL)*, 2024.
5. S-L. Wu, C. Donahue, S. Watanabe, and N. J. Bryan, “Music ControlNet: Multiple Time-varying Controls for Music Generation.” *IEEE Trans. on Audio, Speech, and Lang. Proc. (TASLP)*, 2023.
4. J. Casebeer, N. J. Bryan, P. Smaragdis, “MetaAF: Meta-learning for Adaptive Filtering.” *IEEE Transactions on Audio, Speech, and Language Processing (TASLP)*, 2023.
3. C. J. Steinmetz, N. J. Bryan, J. D. Reiss. “Style Transfer of Audio Effects with Differentiable Signal Processing.” *Journal of Audio Engineering Society (JAES)*, 2022.
2. Z. Tang, N. J. Bryan, D. Li, T. Langlois, D. Manocha, “Scene-Aware Audio Rendering via Deep Acoustic Analysis.” *IEEE VR Journal Papers*, 2020.
1. T. Quirino, M. Kubat, N. J. Bryan. “Instinct-Based Mating in Genetic Algorithms Applied to the Tuning of 1-NN Classifiers.” *IEEE Trans. on Knowledge and Data Engineering*. December 2010.

CONFERENCE PUBLICATIONS (PEER-REVIEWED)

43. F-D. Tsai, S-L. Wu, G. Zhu, J.-P. Caceres, N. J. Bryan. “MuseDuo: Unified Audio-MIDI Music Generation, Separation, Transcription, Rendering, and Editing.” Under review, 2026.
42. J. Mahajan, D. Kim, G. Zhu, J.-P. Caceres, N. J. Bryan. “MusicMorpher: Music Morphing via Style Tokens.” Under review, 2026.
41. J. Yi, M. Kim, N. J. Bryan, D. Kim. “Asymmetric Diffusion Transformer: Representation Supervision via Training-Time Queries.” Under review, 2026.
40. J. Wu, Z. Wang, M. Ostendorf, G. Zhu, S. H. Lee, N. J. Bryan. “Text-to-music generation, structured captions, prompt optimization, text-audio alignment.” Under review, 2026.
39. Y. Wang, S. Agarwal, Z. Jin, N. J. Bryan, Q. Yan, J. Collomosse, V. Asnani. “HierMark: High-Capacity Audio Watermarking with Hierarchical Time-Frequency Embedding.” Under review,

2026.

38. Y-B. Lin, J. Casebeer, L. Mai, A. Mahapatra, G. Bertasius, N. J. Bryan. "V2M-Zero: Zero-Shot Time-Aligned Video-to-Music Generation." *Neural Information Processing Systems (NeurIPS)*, 2026.
37. S. Kumar, P. Seetharaman, K. Chen, O Nieto, J. Su, Z. Wang, R. Kumar, D. Manocha, N. J. Bryan, Z. Jin, J. Salamon. "Timestamped Audio Captioning." *Neural Information Processing Systems (NeurIPS)*, 2026.
36. I. Bukey, Z. Wang, C. Donahue, N. J. Bryan. "Rethinking Music Captioning with Music Metadata LLMs." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
35. J. Casebeer, G. Zhu, Zhepei Wang, N. J. Bryan. "A Generative-first Neural Audio Autoencoder." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
34. S-L. Wu, G. Zhu, J-P. Caceres, C-Z. Huang, N. J. Bryan. "Stemphonic: All-at-once Flexible Multi-stem Music Generation." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2026.
33. Z. Novack, G. Zhu, J. Casebeer, J. McAulay, T. Berg-Kirkpatrick, N. J. Bryan, "Presto! Distilling Steps and Layers for Accelerating Music Generation." *International Conference on Learning Representations (ICLR) (Spotlight, top 5%)*, 2025.
32. Z. Novack, J. McAulay, T. Berg-Kirkpatrick, N. J. Bryan, "DITTO-2: Distilled Diffusion Inference Time T-Optimization for Music Generation." *International Society of Music Information Retrieval (ISMIR)*, 2024.
31. Z. Novack, J. McAulay, T. Berg-Kirkpatrick, N. J. Bryan, "DITTO: Diffusion Inference-Time T-Optimization for Music Generation." *International Conference on Machine Learning (ICML) (Oral, top 1.5%)*, 2024.
30. J. Wu, J. Casebeer, N. J. Bryan, P. Smaragdis, "Meta-learning for Adaptive Filters with Higher-order Frequency Dependencies." *IEEE International Workshop on Acoustic Signal Enhancement (IWEANC)*, 2022.
29. H. Yang, S. Firodiya, N. J. Bryan, M. Kim. "Don't Separate, Learn to Remix: End-to-End Neural Remixing with Joint Optimization," *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2022.
28. M. Won, J. Salamon, N. J. Bryan, G. J. Mysore, X. Serra, "Emotion Embedding Spaces for Matching Music to Stories." *International Society for Music Information Retrieval Conference (ISMIR)*. Online, 2021. **(Best student paper award)**
27. J. Salamon, O. Nieto, N. J. Bryan. "Deep Embeddings and Section Fusion Improve Music Segmentation." *International Society for Music Information Retrieval Conference (ISMIR)*, 2021.
26. Y. Wang, N. J. Bryan, J. Salamon, M. Cartwright, J. P. Bello. "Who Calls the Shots? Rethinking Few-Shot Learning for Audio." *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021. **(IEEE special best paper award)**.
25. J. Casebeer, N. J. Bryan, P. Smaragdis. "Auto-DSP: Learning to Optimize Acoustic Echo Cancellers." *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021.
24. M. A. Martínez Ramírez, O. Wang, P. Smaragdis, N. J. Bryan. "Differentiable Signal Processing with Black-Box Audio Effects." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2021.

23. Y. Wang, N. J. Bryan, M. Cartwright, J. P. Bello, J. Salamon. "Few-shot Continual Learning for Audio Classification." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2021.
22. M. Morrison, L. Rencker, Z. Jin, N. J. Bryan, J.-P. Caceres, B. Pardo. "Context-aware Prosody Correction for Text-based Speech Editing." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2021.
21. P. Manocha, A. Finkelstein, Z. Jin, R. Zhang, N. J. Bryan, G. J. Mysore. "A Differentiable Perceptual Audio Metric Learned from Just Noticeable Differences." *Interspeech*, 2020. **(Best student paper finalist)**
20. M. Morrison, Z. Jin, J. Salamon, N. J. Bryan, G. J. Mysore. "Controllable Neural Prosody Synthesis." *Interspeech*, 2020.
19. J. Lee, N. J. Bryan, J. Salamon, Z. Jin, J. Nam. "Metric Learning vs Classification for Disentangled Music Representation Learning." *International Society for Music Information Retrieval (ISMIR)*, 2020.
18. Y. Wang, J. Salamon, M. Cartwright, N. J. Bryan, J. P. Bello. "Few-shot Drum Transcription in Polyphonic Music." *International Society for Music Information Retrieval (ISMIR)*, 2020.
17. J. Lee, N. J. Bryan, J. Salamon, Z. Jin, J. Nam. "Disentangled Multidimensional Metric Learning for Music Similarity." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
16. Y. Wang, J. Salamon, N. J. Bryan, J.-P. Bello. "Few-Shot Sound Event Detection." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020. **(Press)**
15. S. I. Mimilakis, N. J. Bryan, P. Smaragdis. "One-Shot Parametric Audio Production Style Transfer with Application to Frequency Equalization." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
14. N. J. Bryan. "Impulse Response Data Augmentation and Deep Neural Networks for Blind Room Acoustic Parameter Estimation." *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
13. N. J. Bryan, G. J. Mysore, G. Wang. "ISSE: An Interactive Source Separation Editor." *Conference on Human Factors in Computing Systems (CHI)*, 2014.
12. N. J. Bryan. "Interactive Sound Source Separation." *Ph.D. Thesis*, Stanford University, 2014. **Audio Eng. Society Graduate Student Design Gold Award (2013).**
11. N. J. Bryan, G. J. Mysore, G. Wang. "Source Separation of Polyphonic Music with Interactive User-Feedback on a Piano-Roll Display." *International Society for Music Information Retrieval Conference (ISMIR)*, 2013.
10. N. J. Bryan, G. J. Mysore. "An Efficient Posterior Regularized Latent Variable Model for Interactive Sound Source Separation." *International Conference on Machine Learning (ICML)*, 2013. **(Oral)**
9. N. J. Bryan, G. J. Mysore. "Interactive Refinement of Supervised and Semi-Supervised Sound Source Separation Estimates." *IEEE Int. Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2013.
8. N. J. Bryan, J. Herrera, G. Wang. "User-Guided Variable-Rate Time-Stretching Via Stiffness Control." *International Conference on Digital Audio Effects (DAFx)*, 2012.
7. N. J. Bryan, P. Smaragdis, G. J. Mysore. "Clustering and Synchronizing Multi-Camera Video via Landmark Cross-Correlation." *IEEE International Conference on Acoustics, Speech, and Signal*

Processing (ICASSP), 2012.

6. N. J. Bryan, G. Wang. "Musical Influence Network Analysis and Rank of Sample-Based Music." *International Society for Music Information Retrieval Conference (ISMIR)*, 2011.
5. N. J. Bryan, G. Wang. "Two Turntables and a Mobile Phone." *International Conference on New Interfaces for Musical Expression (NIME)*, 2011. **(Press)**
4. K. Lee, N. J. Bryan, J. S. Abel. "Approximating Measured Reverberation Using a Hybrid Fixed/Switched Convolution Structure." *International Conference on Digital Audio Effects DAFx*, 2010.
3. N. J. Bryan, J. Herrera, J. Oh, G. Wang. "MoMu: A Mobile Music Toolkit." *International Conf. on New Interfaces for Musical Expression (NIME)*, 2010. **(Press)**
2. J. Oh, J. Herrera, N. J. Bryan, G. Wang. "Evolving The Mobile Phone Orchestra." *International Conference on New Interfaces for Musical Expression (NIME)*, 2010.
1. G. Wang, N. Bryan, J. Oh, R. Hamilton. "Stanford Laptop Orchestra (SLORK)." *International Computer Music Conference (ICMC)*, 2009. **(Press)**

OTHER PUBLICATIONS

7. J. Casebeer, N. J. Bryan, P. Smaragdis, "Scaling Up Adaptive Filter Optimizers." *arXiv*, 2024.
6. M. Morrison, Z. Jin, N. J. Bryan, J.-P. Caceres, B. Pardo, "Neural Pitch-Shifting and Time-Stretching with Controllable LPCNET." *arXiv*, 2021.
5. N. J. Bryan, G. J. Mysore. "Interactive User-Feedback for Sound Source Separation." *International Conference on Intelligent User-Interfaces (IUI) Workshop on Interactive Machine Learning (Extended Abstract/Workshop Paper)*. March 2013.
4. J. S. Abel, N. J. Bryan, T. Skare, M. Kolar, P. Huang, D. Mostowfi, J. O. Smith III. "A Configurable Microphone Array with Acoustically Transparent Omnidirectional Elements." *In Audio Engineering Society Convention*. October 2009.
3. N. J. Bryan, M. A. Kolar, J. S. Abel. "Impulse Response Measurements in the Presence of Clock Drift." *In Audio Engineering Society Convention*. November 2010.
2. N. J. Bryan, J. S. Abel. "Methods For Extending Room Impulse Responses Beyond Their Noise Floor." *In Audio Engineering Society Convention*. November 2010.
1. J. S. Abel, N. J. Bryan, et al. "On Estimating Room Impulse Responses from Recorded Balloon Pops." *In Audio Eng. Society Convention*. November 2010.

AWARDS

- Adobe Impact Award, 2025.
- IEEE Senior Member, 2024.
- Adobe Distinguished Inventor, 2023.
- Adobe Intern Showcase 1st Prize (Mentor), 2023.
- Best Student Paper Award (mentor), ISMIR, 2021.

- IEEE Best Audio Few-shot Learning Paper, IEEE WASPAA, 2021.
- Best Student Paper Finalist, Interspeech, 2020.
- Best Reviewer Award, ISMIR, 2020.
- Gold Award, AES Graduate Student Design, 2013.
- U of Miami-FL, Summa cum laude honors, General honors, Dept. of EE honors 2007.
- U of Miami-FL, Tzay Y. Young Best EE Senior Design Award, U of Miami-FL, 2007.
- Frost Wind Ensemble Performance, Carnegie Hall, 2007.
- U of Miami-FL, Research & Creativity Fair, Undergrad 2nd Place Winner, 2006.

OPEN-SOURCE SOFTWARE

- **Convmelspec** – Convertible Melspectrograms via 1D Convolutions (<https://github.com/adobe-research/convmelspec>), released 2022.
- **MetaAF** – Control adaptive filters with neural networks (<https://github.com/adobe-research/MetaAF>), released 2022.
- **DeepAFx-ST** – Style transfer of audio effects with differentiable signal processing (<https://github.com/adobe-research/DeepAFx-ST>), released 2022.
- **DeepAFx** – Third-party audio effects plugins as differentiable layers within deep neural networks (<https://github.com/adobe-research/DeepAFx>), released 2021.
- **ISSE** – Cross-platform interactive source separation editor via drawing and painting (<http://isse.sourceforge.net>). 82,000+ downloads, 120+ countries. released 2013.

PATENTS SUBMITTED

- Z. Novack, G. Zhu, J. Casebeer, N. J. Bryan, “Accelerated Diffusion Transformer Audio Generation, 2025.
- G. Zhu, J.P. Caceres, N. J. Bryan. “MusicHiFi: An Efficient High-Fidelity Vocoder”, 2025.
- J. Casebeer, N. J. Bryan, “Meta-Learning for Adaptive Filters”, 2025.
- S-L. Wu and N. J. Bryan, “Music Generation with Time Varying Controls”, 2024.
- C. J. Steinmetz, N. J. Bryan, Simon Jenni, and John Collomosse, “General Purpose Audio Fingerprinting”, 2023
- J. Salamon, J.-P. Caceres, G. Zhu, N. J. Bryan. “A Pipeline for Filler Word Detection and Classification.” Submitted by Adobe, August 2022.
- N. J. Bryan, P. Smaragdis. “Method for Learning to Optimize Signal Processing Algorithms.” Submitted by Adobe, April 2022.
- J. Salamon, O. Nieto, N. J. Bryan, “Multi-Level Audio Segmentation Using Deep Embeddings.” Submitted by Adobe, Oct. 2022.
- N. J. Bryan and J. Salamon, “Section-Based Music Similarity Searching.” Submitted by Adobe, Oct. 2022.

PATENTS ISSUED

14. M. Morrison, J.P. Caceres Chomali, Z. Jin, N. Bryan, B.A. Pardo — Neural Pitch-Shifting And Time-Stretching | US11915714 | Issued 2/27/2024
13. M.A. Martinez Ramirez, N.J. Bryan, O. Wang, P. Smaragdis — Deep Encoder For Performing Audio Processing | US11900902 | Issued 2/13/2024
12. M. Morrison, Z. Jin, N. Bryan, J.P. Caceres Chomali, L. Rencker — Context-Aware Prosody Correction Of Edited Speech | US11830481 | Issued 11/28/2023
11. J. Salamon, Y. Wang, N.J. Bryan — Automated Sound Matching Within An Audio Recording | US11501102 | Issued 11/15/2022
10. N. Bryan — Voice Recordings Using Acoustic Quality Measurement Models And Actionable Acoustic Improvement Suggestions | US11462236 | Issued 10/4/2022
9. J. Lee, N.J. Bryan, J. Salamon, Z. Jin — Searching For Music | US11461649 | Issued 10/4/2022
8. Z. Tang, T. Langlois, N. Bryan, D. Li — Rendering Scene-Aware Audio Using Neural Network-Based Acoustic Analysis | US11190898 | Issued 11/30/2021
7. S.I. Mimilakis, P. Smaragdis, N. Bryan — Audio Production Assistant For Style Transfers Of Audio Recordings Using One-Shot Parametric Predictions | US11082789 | Issued 8/3/2021
6. N. Bryan — Generating Synthetic Acoustic Impulse Responses From An Acoustic Impulse Response | US11074925 | Issued 7/27/2021
5. N.J. Bryan, Q. Yang, V. Iyengar — Mitigating Noise in Audio Signals | Apple | 2021
4. N.J. Bryan, V. Iyengar — Speech Enhancement for an Electronic Device | US20190272842 | Apple | 2020
3. N.J. Bryan, V. Iyengar, A. Lindahl — Transparent Near-End User Control Over Far-End Speech Enhancement Processing | US20190156847A1 | Apple | 2019
2. N.J. Bryan, V. Iyengar — System and Method of Noise Reduction for Mobile Device | US20180350381A1 | Apple | 2019
1. N.J. Bryan, P. Smaragdis, G.J. Mysore — Clustering and Synchronizing Content | US8924345B2 | Adobe | 2014

PhD THESIS COMMITTEES

- Zachary Novack – UCSD, 2026.
- Ge Zhu – University of Rochester, 2025.
- Eloi Moliner Juanpere – Alto University (external examiner), 2025.
- Jonah Casebeer – University of Illinois Urbana-Champaign, 2023
- Yu Wang – New York University, 2023
- Jongpil Lee – Korea Advanced Institute for Science and Technology, 2020

TECHNICAL COMITTEES

- IEEE Audio and Acoustic Signal Processing (AASP) Technical Committee (2023-2025)
- IEEE Audio and Acoustic Signal Processing (AASP) Technical Committee (2020-2022)

AREA CHAIR, META REVIEWER, SESSION CHAIRS

- ICASSP 2027 Area Chair
- ISMIR 2026 Meta Reviewer
- ACM Creativity and Cognition Conference – Meta Reviewer (6)
- ICASSP 2026 AASP TC Area Chair – Audio Security (14)
- ISMIR 2025 Meta Reviewer (5)
- IEEE WASPAA 2025 Area Chair (5)
- IEEE ICASSP 2025 AASP TC Area Chair—Data & Open Source (16)
- ISMIR 2024 Meta-Reviewer (5)
- IEEE ICASSP 2024 AASP TC Area Co-Chair– Music Signal Analysis, Proc, and Synth (38)
- IEEE AASP TC – EDICS Vice-Chair 2025
- IEEE WASPAA 2023 General Co-Chair
 - Co-led review process transition from single- to double-blind
 - Co-led 2-3x increase in corporate sponsorship donations
 - Co-led 10x increase in student travel grants
 - Led first individual donation campaign w/IEEE Foundation
 - Co-led 2-2.5x increase in student attendance at WASPAA
- IEEE ICASSP 2023 Area Chair – Music Signal Analysis, Processing, and Synthesis (34)
- ISMIR 2022 Meta Reviewer (3)
- IEEE ICASSP 2022 Area Chair – Music Signal Analysis, Processing, and Synthesis (41)
- ISMIR 2021 Meta Reviewer (3)
- IEEE WASPAA 2021 Area Chair – Music Signal Analysis, Processing, and Synthesis (12)
- IEEE ICASSP 2021 AASP – Deep Learning for Music Signal Processing Session Chair
- IEEE ICASSP 2021 AASP – Music Signal Analysis, Processing & Synthesis Area Chair (24)
- IEEE ICASSP 2020 AASP – Music Information Retrieval Session Chair

ACADEMIC REVIEWING

- 2026 – ACM C&C (5), ECCV (1), NeurIPs (1)
- 2025 – WASPAA (5), ISMIR (5), TMLR (1)
- 2024 – TASLP (1), SIGGRAPH (1)
- 2023 – TASLP (3), SPL (1)
- 2022 – IEEE SPL (1), IEEE TASLP (3)
- 2021 – IEEE TASLP (2), ISMIR reviews (3), Pattern Recognition (1)
- 2020 – ICASSP (12), SIGGRAPH (1), IEEE TASLP (3), ISMIR (4, **Best Reviewer**)
- 2019 – ICASSP (10), WASPAA (6), IEEE TSP (1), IEEE SPL (1), Stanford HAI Blog (1)

- 2018 – ICASSP (3), EUSIPCO (5), NIME
- 2017 – ICASSP, WASPAA, EUSIPCO, ACM Trans. on Interactive Int. Systems, NIME
- 2016 – ICASSP, WASPAA, EUSIPCO, ACM Trans. on Interactive Int. Systems, NIME
- 2015 – ICASSP, WASPAA, EUSIPCO, JASA, NIME, IEEE Journal Topics Signal Proc.
- 2014 – ICASSP, Elsevier Signal Processing, NIPS, NIME, Computer Communications

SELECTED PRESS

- [Now in Beta: Generate Music in Premiere](#) (8/20/26)
- [Adobe Firefly expands its creative AI studio: generate music, speech, & sfx in one place](#) (8/20/26)
- Generate Soundtrack, Adobe News + 2x Keynote Demos (10/28, 2025)
<https://news.adobe.com/news/2025/10/adobe-max-2025-firefly>
- Project Music GenAI Control, Adobe News (2/2024)
<https://blog.adobe.com/en/publish/2024/02/28/adobe-research-audio-creation-editing>
- MAX Sneaks – “SoundSeek.” (11/19). <https://www.youtube.com/watch?v=ebAlnW3Xs2k>
- MAX Sneaks – “Audio Layers.” Adobe TV (05/13).
http://www.youtube.com/watch?feature=player_embedded&v=nyODK-J9keo
- MAX Sneaks – “Automatic Synchronization of Crowd Sourced Videos.” Adobe TV (10/11).
<https://www.youtube.com/watch?v=dhyvzOJBhzc>
- “Smartphone turntables put new spin on DJing.” NewScientist (05/11).
<https://tinyurl.com/newscientist-mophodj>
- "Air Instruments only." *The Stanford Daily* (10/10). <http://www.stanforddaily.com/2010/10/28/air-instruments-only>
- "From Pocket to Stage, Music in the Key of iPhone." NY Times (*Front Page* 12/04/09).
http://www.nytimes.com/2009/12/05/technology/05orchestra.html?_r=2&hpw
- “Make-TV: Computer Making Music (Episode 9).” PBS (02/09).
<https://www.youtube.com/watch?v=R2OF2OLaHdw>
- “Stanford Students invent new music.” Local ABC TV (SF), Drive to Discover (04/08).
http://abclocal.go.com/kgo/story?section=news/drive_to_discover&id=6064990
- “Rhythm and Cubes.” Stanford Magazine (07/08).
<http://www.stanfordalumni.org/news/magazine/2008/julaug/red/cubeats.html>

MENTORING

- Jayeon Yi (2026) – UIUC
- Junkai Wu (2026) – UW
- Jay Mahajan (2026) – UIUC
- Fang-Duo Tsai (2026) – National Tawain University
- Irmak Bukey (2025) – CMU
- Nithya Shikarpur (2025) – MIT

- Dimitrios Bralios (2025) – UIUC
- Yan-Bo Lin (2025) – UNC
- SeungHeon Doh (2024) – KAIST
- Yatong Bai (2024) – University of California Berkeley
- Zachary Novack (2023-2024) – University of California San Diego
- Ge Zhu (2023) – University of Rochester
- Shih-Lun Wu (2023, 2025) – CMU, MIT
- Jonah Casebeer (2020-2022) – University of Illinois Urbana-Champaign
- Christian Steinmetz (2021, 2022) – Queen Mary University of London
- Jack Atherton (2021) – Stanford University
- Gabriel Zalles (2021) – University of California San Diego
- Amber Blue (2021) – Georgia Tech
- Minz Won (2020-2021) – Universitat Pompeu Fabra
- Marco Antonio Martínez Ramírez (2020) – Queen Mary University of London
- Yu Wang (2019-2021) – New York University
- Jongpil Lee (2019-2020) – Korea Advanced Institute for Science and Technology
- Stylianos Mimilakis (2019) – Technical University of Ilmenau
- Lucas Rencker (2019) – University of Surrey
- Zhenyu Tang (2019) – University of Maryland
- Maxwell Morrison (2019-2021) – Northwestern University

SELECTED TALKS & WORKSHOPS

- “The Grand Design Challenge of Music GenAI”, Conversational AI Reading Group at Mila, May 2026.
- “Creative AI: Music, Authorship, and Algorithmic Culture,” UCSD Design@Large Speaker Series Invited Speaker (of three guests), April 2026.
- “Beyond Text to Music” Keynote Talk, [GenDA](#) Workshop, ICASSP 2025.
- “My Career Journey”, University of Miami Music Engineering Forum, 2025.
- “Learning to Control Signal Processing Algorithms with Deep Learning”, *Women in Music Information Retrieval Workshop*, 2021.
- “ISSE: An Interactive Machine Learning Workflow.” (Workshop Demo) NIPS *Workshop on Machine Learning Open-Source Software*. December 2013.
- “ISSE: An Interactive Source Separation Editor.” (Two-part Talk) *CCRMA DSP Seminar*, Stanford University. October and November 2013.
- “Audio Source Separation.” (Workshop Speaker w/Gautham J. Mysore, Pierre Leveau, Derry FitzGerald, Elias Kokkinis), *Audio Eng. Society Convention*. October 2013.
- “Source Separation Tutorial Mini-Series.” (Three-part Talk w/Dennis Sun and Eunjoon Cho)

CCRMA DSP Seminar, Stanford University, April 2013.